



RKDF UNIVERSITY RANCHI

M.Sc. Botany

Branch	Subject title	Subject Code
M.Sc. Botany	Bacteriology, Virology & General Microbiology	PBN101

Unit-1

Bacterial taxonomy; Identification of bacteria, General characters of Rickettsia and Chlamydia, Diseases caused by Rickettsia and Chlamydia, Mode of nutrition in bacteria; autotrophy, heterotrophy, symbiosis.

Unit-2

General account of sterilization culture media, pure culture techniques; A general idea about bacterial toxins and enzymes; Bacteriophage; Bacterial diseases: caused by Escherichia coli, Shigella.

Unit-3

General properties and evolution of viruses; Cultivation of virus and viral assay; Transmission of plant viruses and control measures, Oncogenic viruses and tumorigenesis; Viral diseases: Encephalitis, hepatitis AIDS and Rabies

Unit-4

Biological nitrogen fixation: symbiotic and non symbiotic nitrogen - fixation; Fermentation technology: principle and types of fermentation, Microbial degradation of pesticides and hydrocarbons; Mycoplasma: general account and important diseases caused by them

Unit-5

Microbial conversion of waste product with particular reference to alcohol and biogas, General account of Immunity, properties of antigens and antibodies, Allergy and types of allergies, Mycotoxins and their harmful effects.



RKDF UNIVERSITY RANCHI

M.Sc. Botany

Branch	Subject title	Subject Code
M.Sc. Botany	Biology And Diversity of Fungi And Plant Pathology	PBN102

Unit-1

Recent trends on the classification of fungi with reference to morphological and paramorphological criteria; Comparative study of following sub-division; Mastigomycotina: Albugo, Peronospora, Plasmopora.

Unit-2

Comparative study of following sub-division; Zygomycotina: Mucor, Rhizopus, Syncephalastrum, Ascomycotina: Tapharina, Emericella, Penicillium, Chaetomium, Morchella.

Unit-3

Comparative study of following sub-division; Basidiomycotina: Puccinia, Melampsora, Ustilago, Polyporus, Cyathus, Deuteromycotina: Fusarium, Cercospora, Colletotrichum. Mushroom cultivation: Mycorrhizal application in agriculture and forestry; Fungal cytology and genetics: Heterothallism, heterokaryosis, parasexual cycle, mutation.

Unit-4

Symptomatology in fungal, bacterial and viral infection of plants, Etiology and control of the following crop diseases

1. Paddy: paddy blast, paddy blight
2. Wheat: Black stem rust, Bunt of wheat
3. Bajra: green ear and Ergot
4. Sugarcane: Red rot disease of sugarcane.
5. Ground nut: Tikka disease
6. Maize Smut

Unit-5

Role of enzyme and toxins in pathogenesis; Disease control by physical, chemical and biological methods, resistant varieties; Crop rotation, plant quarantines, seed certification.



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M.Sc. Botany

Branch	Subject title	Subject Code
M.Sc. Botany	Biology And Diversity of Algae, Bryophytes And Lichens	PBN103

Unit-1

Comparative survey of important systems of classification of algae; Criteria for algal classification and modern trends; Diagnostic features of algal phyla, range of thallus and reproductive diversity; life history patterns, parallelism and evolution.

Unit-2

Comparative account of algal pigments ; light microscopic structure, ultra structure and function of Cell wall, flagella, chloroplast, pyrenoids and eyespots and their importance in taxonomy. Study of Cyanophyta, Chlorophyta, Xanthophyta, Bacillariophyta, Phaeophyta and Rhodophyta up to the ordinal level with reference to the following genera: Anabaena, Gonium, Chlorella, Enteromorpha, Bulbochaete, Closterium, Acetabularia, Nitella, Botrydium, Navicula, Cyclotella, Batrachospermum and Gracillaria.

Unit-3

General characteristics of the division: Diophyta, Chrysophyta and Cryptophyta. Distribution of algae in soil, fresh water and marine environment, role of algae in soil fertility, productivity in fresh water and marine environment algae role in fisheries, algae in symbiotic association, algae in polluted habitats, algae as indicator of pollution, fossil algae, algae in biotechnology.

Unit-4

Origin of Bryophytes: Primitive vs. advanced characters, derived features: evolutionary lines, Classification. Comparative morphological, anatomical and cytological studies of gametophyte and sporophytes of Calobryales, Jungermanniales, Sphaerocarpaceae, Marchantiales, Takakiales, Sphagnales, Andreales and Bryales

Unit-5

Experimental studies in Bryophytes, Spore germination, Protonemal differentiation, bud formation, Parthenogenesis, apogamy, apospory and regeneration; Bryogeographical regions of India with reference to central India; Lichens: General account, structure and reproduction.



RKDF UNIVERSITY RANCHI

M.Sc. Botany

Branch	Subject title	Subject Code
M.Sc. Botany	Crop Pathology	PBN104

Unit-1

Introduction to crop pathology: Classification of plant diseases based on;

- (a) Major causal agents - biotic and abiotic,
- (b) General symptoms.

Unit-2 Process of infection and pathogenesis:

- (a) Penetration and entry of pathogen into host tissue – mechanical, physiological and enzymatic.
- (b) Host-parasite interaction, enzymes and toxins in pathogenesis.

Unit-3

Defense mechanism in plants: Pre-existing structural and biochemical defense mechanisms, lack of essential nutrients. Induced structural and biochemical defense mechanisms, inactivation of pathogen enzymes and toxins, altered biosynthetic pathways.

Unit-4

Transmission of plant disease: Spread and transmission of plant diseases by wind, water, seeds and vectors. Plant disease management: Exclusion, eradication and protection. Chemical means of disease control – common fungicides, antibiotics and nematicides. Biological means of disease control. Biotechnological approaches to disease resistance: Fungi in agricultural biotechnology, control of fungal plant pathogens by mycofungicides; Transgenic approaches to disease resistance.

Unit-5- Major diseases in plants

- (a) Cereals: Rice - blast disease, bacterial blight; Wheat - black rust disease.
- (b) Vegetables: Chilly - leaf spot; Ladies finger - vein clearing disease.
- (c) Fruits: Banana - bacterial leaf blight, leaf spot; Mango - Anthracnose; Citrus - bacterial canker; Papaya – mosaic.
- (d) Spices: Ginger - rhizome rot; Pepper - quick wilt; Cardamom - marble mosaic disease.
- (e) Oil seeds: Coconut - grey leaf spot, bud rot disease.
- (f) Rubber yielding: Hevea brasiliensis - abnormal leaf fall, powdery mildew.
- (g) Sugar yielding: Sugarcane - red rot; root knot nematode.
- (h) Cash crops: Arecanut - nut fall disease.
- (i) Beverages: Tea - blister blight; Coffee - rust.



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References

1. K S Bilgrami, H C Dube. A text book of modern plant pathology.
2. Gareth Johnes. Plant pathology: principles and practice.
3. R S Mehrotra. Plant Pathology. 4. M N Kamat. Practical plant pathology.



RKDF UNIVERSITY RANCHI

M.Sc. Botany

Branch	Subject title	Subject Code
M.Sc. Botany	Practical-I Lab	PBN155

1. Collection and study of algae mentioned in theory, identification up to generic level using algal monographs.
2. Preparation of synthetic medium and cultivation of algae, unialgal and axenic culture and their maintenance.
3. Collection, preservation of algal herbarium (10 specimens).
4. Preparation of pigments.
5. Staining techniques of cytology studies.
6. Study of diseased specimens of plants with reference to symptomatology.
7. Isolation, purification and single spore culture of pathogens.
8. Measurement of the activity of enzymes of fungal pathogens: Cellulase, Pectinases.
9. Laboratory testing of fungicides (systemic and non-systemic) against pathogenic fungi. Preparation of culture media.
10. Preparation of culture media.
11. Isolation of Bacillus and Rhizobium spp from soil and nodules.
12. Various methods of bacterial staining to study cell wall, endospore, capsule and flagella.
13. Identification of important genera by using biochemical tests: Escherichia, Azatobactor, Staphylococcus, Bacillus, Pseudomonas, Rhizobium, Streptomyces, Xanthomonas.
14. Construction of bacterial growth curve.
15. Quantitative estimation of bacteria in milk
16. Make suitable micropreparations and identify the diseases mentioned with due emphasis on symptoms and causative organisms.
17. Isolation of pathogens from diseased tissues (leaf, stem and fruit) by serial dilution method.



RKDF UNIVERSITY RANCHI

M.Sc. Botany

Branch	Subject title	Subject Code
M.Sc. Botany	Angiosperm Anatomy, Embryology and Palynology	PBN201

Unit-1

Origin, growth, differentiation and ultra structure of cell and tissue, fine structure of plasmodesmata, microtubules, microfibrils and secondary structure; Apical, lateral and intercalary meristems- their ultra structure and histochemistry, organogenesis; Ontogeny, phylogeny, ultra structure and function of primary and secondary xylem; wood anatomy; Ontogeny, phylogeny, ultra structure and function of primary and secondary phloem; Structure variability in leaves, leaf histogenesis, leaf meristem, origin, development ultra structure of trichomes and stomata.

Unit-2

Nodal anatomy-nodal types and evolutionary consideration, Vascular cambium vs. cork cambium factors controlling their activity, periderm, lenticles, abscission, wound healing; Anatomy of monocotyledons and dicotyledonous seed and fruits, seed appendages, their anatomy structure and function; Anatomy in relation to taxonomy; Contemporary plant anatomy: current trends and prospects.

Unit-3

Microsporangium- structure and function of wall layers, ultra structure change in tapetum and meiocytes during Microsporogenesis, role of tapetum, pollen development, anther culture and haploid plants; Pollen wall morphogenesis-microspore pollen mitosis; division of generative cell; pollen fertility and sterility; pollen storage viability and germination; Megasporogenesis, various types of embryosacs, their development and fertilization.

Unit-4

Embryology and taxonomy; diagnostic embryological characters, primitive and advanced characters, comparative embryology of hybrids dysfunction of endosperm, arrested development of embryo.

Unit-5

Development and evolution of pollen types; stereo and ultrastructure of exine, apertures, furrow. Palynology and taxonomy; Aerobiology and its application. Aeropalynology, methods of aerospora survey and analysis, pollen allergy and pollen calendars system approach for allergy. Mellitopalynology: general account Paleopalynology: role in coal and oilgenesis.



RKDF UNIVERSITY RANCHI

M.Sc. Botany

Branch	Subject title	Subject Code
M.Sc. Botany	Angiosperm Morphology and Taxonomy	PBN202

Unit-1

General concept of morphology, origin and evolution of flower. Co-evolution of flower, vis a vis pollinators; Origin and evolution of polypetal, sympetal, apetal; monoc, dioc. Monocot flower.

Unit-2

Stamens: origin and evolution from foliar to reduced condition, extension of connective beyond anthers; mono di and polyadephy; nectaries and nectar; Carpels: evolution, conduplicate, involute and other types, validity of the concept of foliar origin of carpel: alternative concepts and approaches ; specialized carpels; poly and syncarpy; superior, semiinferior and inferior ovary; appendicular and receptaclar concepts; evolution of types of placentations.

Unit-3

Role of floral anatomy in interpreting the origin and evolution of a flower and floral parts. Floral anatomy and taxonomy; Experimental study on flower.

Unit-4

Botanical exploration-historical perspective, brief account of botanical exploration in south east Asia with special reference to India. Botanical survey of India, its organization and role; Principles of plant classification with emphasis on modern tools of taxonomy; molecular systematics; utility of taxonomy; biosystematics; Phylogenetic systems of classification ; Cronquist, Takhtajan, AGP III

Unit-5

Botanical nomenclature! ICBN, principles, articles, recommendation and amendments of code; Familiarity with botanical literature, monographs, icons and floras, important periodicals with emphasis on Indian floristics, methods of literature consultation; Threat assessment, different categories of threat, IUCN, Red Data Book. Important threatened plants of India.



RKDF UNIVERSITY RANCHI

M.Sc. Botany

Branch	Subject title	Subject Code
M.Sc. Botany	Water Relations, Growth and Development	PBN203

Unit-1

Water relations of plants: Unique physio-chemical properties of water, chemical potential , water potential. Apparent free space, bulk movement of water, soil plant atmosphere, contuum (SPAC), stomatal regulation of transpiration, hormonal and energy dependent hypothesis; Inorganic nutrition, physicochemical aspects of solute transport, diffusion and facilitated diffusion, passive and active transport; Nernst equation and Donnan's potential. Role of ATPase as a carrier, co-transport (symport) and counter transport (antiport). Ion channels, role of calmodulin. Importance of foliar nutrition and use of chelates.

Unit-2

Photosynthesis: Energy pathway in photosynthesis, chloroplast as an energy transducing organelle; Composition and characterization of photo systems, I and II , electron flow through cyclic, non cyclic and pseudo cyclic photophosphorylation. Pathways of CO₂ fixation. Differences between C₃ and C₄ fixation and different kinds of C₄ pathways.

Unit-3

CAM pathway: Occurrence, biological events and adaptive advantage; Photorespiration : Mechanism and regulation of photorespiration; Introductory studies on water stress and its tolerance mechanisms.

Unit-4

Enzymes: Classification, mode of action, Km value; Industrial application, immobilized enzymes, their preparation and application; Enzyme regulation : Competitive and non-competitive , allosteric enzymes.

Unit-5

Chemical control of growth and morphogenesis; Hormonal effects on growth and development; Bioassay of plant growth regulators and mode of action with reference to auxins; Gibberellins, cytokinins, abscisic acid and ethylene; Phytochrome: Chemistry and photo morphogenetic effects and role in flowering; Genetic study of secondary metabolites such as alkaloids (only types of wide occurrence.) Dormancy: Seed and bud dormancy; hormonal regulation.



RKDF UNIVERSITY RANCHI

M.Sc. Botany

Branch	Subject title	Subject Code
M.Sc. Botany	Ecology - Climatology, Soil Science and Autecology	PBN204

Unit-1

Definition, scope and concept of plant ecology; History of ecology and relation of ecology with other disciplines. Principles of ecology; Concept of environment, habitat and ecological niche. The environment we live in.

Unit-2

Light and temperature as ecological factors; Precipitation and Relative Humidity as ecological factors; Measurement and analysis of light, temperature, Precipitation and relation Humidity; Importance of water as an important factor on the life of plants.

Unit-3

Origin, development and formation of soil. Soil profile; Classification of soil; Effects of soil environment of plants; Chief soil types of India.

Unit-4

Biotic components of an ecosystem; Interrelation of various organisms; Population ecology, Natality, Mortality, Age distribution; Concept of carrying capacity.

Unit-5

Morphological, anatomical and physiological relation of plants with their environment; Plant indicators; Ecotypic and Ecadic differentiation; Physical and physiological dryness; Genecology

References:

1. V K Gupta, T S Paul. Fungi and Plant disease.
2. Malhotra, Aggarwal Ashok. Plant Pathology.
3. Rangaswamy, A Mahadevan. Diseases of crop plants in India.
4. B P Pandey. Plant Pathology.
5. George N Agrios (2006). Plant pathology (V Edn). Elsevier Academic Press.



RKDF UNIVERSITY RANCHI

M.Sc. Botany

Branch	Subject title	Subject Code
M.Sc. Botany	Practical –II Lab	PBN255

PRACTICALS

1. Study of physical and chemical characteristics of soil by rapid field test.
2. Determination of pH of water.
3. Determination of dissolved oxygen in water
4. Extraction of amylase and determination of its activity
5. Preparation of cleared whole mounts of floral parts of polypetalae, sympetalae and monocots for vasculature.
6. With the help hand section and dissection prepare longitudinal and transverse sections of flower. Examination of: a. Transmitting tissue/ canal in stigma and style. Use of paraffin method of microtechnique .
7. Acquaintance with ultratome: use of wood microtome and common and anatomy and histochemical methods.
8. Learning techniques of making temporary and permanent microscopic preparation.
9. Knowledge and use of photomicrography in anatomical studies.
10. Knowledge and use of the principles and working of electron microscope
11. Learning to use simple experimental method in anatomical studies.
12. Determination of moisture constant of soil.
13. Determination of pH of water.
14. Determination of dissolved oxygen in water
15. Determination of following data. a. Solar energy b. Atmospheric temperature c. Relative Humidity
16. Determination of soil profile.



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M.Sc. Botany

Branch	Subject title	Subject Code
M.Sc. Botany	Biochemistry and Metabolism in Plants	PBN301

Unit-1

Lipid: General structure, classification; Synthesis of fatty acid; β -Oxidation; Synthesis of carbohydrates from fatty acids; Protein: Amino acid structure and Biosynthesis; protein conformation, protein synthesis.

Unit-2

Secondary plant products: General structure of important phenolic compound groups in plants; Shikimic acid pathway and phenolic compound synthesis; General structure and synthesis of alkaloids from amino acids; Growth regulators: structure and biosynthesis of Ethylene, Jasmonates and Brassinolides.

Unit-3

Cell wall components; Cellulose: structure and model for biosynthesis, structure and operation of the model for synthesis at plasma membrane level; Lignin: structure and biosynthesis in plants. Cyanogenesis: a general account of cyanogenic compounds as glycosides.

Unit-4

DNA and RNA: structure; DNA replication; Transcription in prokaryotes and eukaryotes, transcript modification; Regulation of translation.

Unit-5

Nitrate metabolism in plants; Nitrogen fixation, Nitrogenase system; Ammonium assimilation. GS-GOGAT system;

References :

1. Jaibala, S. and Balakrishnan, G. (1975). A Hand book of common remedies based On Siddha system of Indian medicines. St. Louis Institute Press, Chennai.
2. Natkarni, K.M. (1998). Indian Materia Medica (Vol. I-III). Popular Prakasam, New Delhi.
3. Raychaudri, S.P. (1991). Recent advances in Medicinal, Aromatic and Spice crops (Vol. I). Today & Tomorrow publication, New Delhi.
4. Johnson, T (1999). CRC Ethnobotany desk Reference , CRC press ,New York



RKDF UNIVERSITY RANCHI

M.Sc. Botany

Branch	Subject title	Subject Code
M.Sc. Botany	Plant Breeding	PBN302

Unit-1

Introduction: Objectives of plant breeding, important achievements and future prospects. Genetic variability and its role in plant breeding. Domestication and centers of origin of cultivated plants.

Unit-2

Systems of reproduction in plants: Reproductive systems and pollination control mechanisms; Sexual reproduction - Cross and self pollination; asexual reproduction, Incompatibility and Male sterility, their types.

Unit-3

Hybridization: Hybridization - role and methods, Inter-varietal, inter specific and inter generic crosses. Back-cross breeding. Heterosis, Inbreeding depression.

Unit-4

Breeding for resistance: Breeding for biotic (disease) and abiotic (drought) stresses; loss due to diseases, disease development, disease escape, disease resistance, vertical and horizontal resistances of biotic stress; methods of breeding for disease resistance.

Unit-5

Mutation breeding: Mutagens and crop improvement. Spontaneous and induced mutations, effects of mutation. Physical and chemical mutagens; principles and working of Gamma gardens, methods of mutation breeding, mutations in oligogenic traits, mutations in polygenic traits, limitations of mutation breeding, achievements of mutation breeding. Role of mutations in Plant Breeding; Modern trends in plant breeding.

References

1. Allard R W (1995). Principles of Plant Breeding. John Wiley and Sons, Inc.
2. Ghahal G S and Gosal S S (2002). Principles and procedures of Plant Breeding. Narosa Publishing House.
3. Sharma J R (1994). Principles and practices of Plant Breeding. Tata McGraw-Hill Publishers Company Ltd.
4. Singh B D (1996). Plant Breeding: Principles and methods. Kalyani Publications.



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M.Sc. Botany

Branch	Subject title	Subject Code
M.Sc. Botany	Plant Biotechnology: In Vitro Culture, Genetic Engineering And IPR Issue	PBN303

Unit-1

Concept and scope of Biotechnology; Techniques of tissue culture, cell culture and organ culture; Sterilization, culture media; In-vitro auxorrophs, disease resistance, salt and drought resistance, nutritional quality and herbicide resistance.

Unit-2

Micropropagation; Production of haploids: anther culture and pollen culture Somatic embryogenesis, somaclonal variation; Protoplast culture: isolation, culture and fusion of protoplast; IPR-general idea about patents. Copyright, trademark and geographical indication.

Unit-3

Biotransformation: production of useful compounds through cell culture; factors affecting yield: bioreactors; Strategies of microbial strain improvement; The recombinant DNA concept and principle of cloning; Isolation and purification of DNA.

Unit-4

Restriction endonuclease : properties and types; Blotting southern, northern and western; Selection and screening of recombinant clone; Cloning vehicles salient features: plasmid , cosmid & Tiplasmid.

Unit-5

Single stranded DNA viruses CaMV Lambda phage vectors M13 vectors; Expression vectors; Cloning construction of genomic and DNA libraries; Application of- r- DNA technology in plant improvement.

References:

1. Jaibala, S. and Balakrishnan, G. (1975). A Hand book of common remedies based On Siddha system of Indian medicines. St. Louis Institute Press, Chennai.
2. Natkarni, K.M. (1998). Indian Materia Medica (Vol. I-III). Popular Prakasam, New Delhi.
3. Raychaudri, S.P. (1991). Recent advances in Medicinal, Aromatic and Spice crops (Vol. I). Today & Tomorrow publication, New Delhi.
4. Johnson,T (1999). CRC Ethnobotany desk Reference , CRC press ,New York



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M.Sc. Botany

Branch	Subject title	Subject Code
M.Sc. Botany	HERBAL TECHNOLOGY, INDUSTRIAL MICROBIOLOGY	PBN304

Unit-1

History of Medicinal plants. Traditional Medicinal systems: Ayurvedha, Siddha, Unani and Naturopathy. Definition of Drug - Classification of natural drugs, (Alphabetical, Morphological, Pharmacological, Chemical and Chemo taxonomical). Traditional and Folklore medicines - Native medicine.

Unit -2

Pharmacognosy - Definition and Scope. Drug adulteration, Drug evaluation, Chemical evaluation, Physical evaluation and Biological evaluation. Phytochemical investigations,

Unit -3

Cultivation, collection and preparation of natural drugs - Macroscopic characters (Physical and Organoleptic characters), therapeutical and pharmaceutical uses of the following medicinal plants: Adathoda vasica, Aloe vera, Centella asiatica, Syzygium aramoticum, Datura metel, Piper nigrum, Allium sativum, Azadiracta indica, Ocimum sanctum and Vinca rosea.

Unit –4

Industrial Microbiology- Industrial product of vinegar & citric acid. Industrial product of antibiotics; penicillin and streptomycin Microbes as a source of Single Cell protein (SCP). Mushrooms Dairy product from microorganisms; butter, yogurt and cheese. Biopesticides, Biofertilizer, Biopolymers, Bioremediation.

Unit –5

Ethnobotany – definition-Major tribes of South India and their ethno botanical and ethnobiological heritage. Ethno Medicines. Ethnobotany and conservation of plants with special reference to India – mythology and conservation of ecosystems, conservation of selected plant species: sacred groves, forestry and unique ecosystems and their ethnobiological values, plants and animals in art, tradition and ethnography: Ethnobotanical field methods.

Text books 1. John Jothi Prakash, E. (2003). Medicinal Botany and Pharmacognosy. Page 41 of 96 JPRPublication, Vallioor, Tirunelveli.



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2. Gokhale, S.B., Kokate, C.K. and Purohit, A.P. (2003). Pharmacognosy. Nirali Prakashan, Pune.
3. Prajapathi, Purohit, Sharma and Kumar. (2003). A Hand book of Medicinal plants. Agrobios Publications, Jodhpur.



RKDF UNIVERSITY RANCHI

M.Sc. Botany

Branch	Subject title	Subject Code
M.Sc. Botany	Practical – III Lab	PBN355

Practicals:

1. Estimation of total nitrogen by kjaldahl method.
2. Principles of colorimetry, spectrophotometry and fluorimetry.
3. Estimation of titrable and total acidity.
4. Estimation of protein by Biuret and Lowry's method.
5. Determination of probability of tossing for one coin.
6. Determination of probability for the throw of dice.
7. Determination of probability for tossing of two coins.
8. X 2 test as applied to the result of above three experiments.
9. Selection of salt tolerance / amino acid analogue resistance through cell culture.
10. Isolation and culture of protoplast.
11. Isolation and screening of industrially important microorganism.
12. Isolation of plant DNA, plasmid DNA, bacteriophage DNA.



RKDF UNIVERSITY RANCHI

M.Sc. Botany

Branch	Subject title	Subject Code
M.Sc. Botany	Dissertation	PBN466

Overall project dissertation may be evaluated under the following heads:

- ☐ Motivation for the choice of topic
- ☐ Project dissertation design
- ☐ Methodology and Content depth
- ☐ Results and Discussion
- ☐ Future Scope & References
- ☐ Presentation style
- ☐ Viva-voce

PROJECT WORK/ DISSERTATION/ PAPER PRESENTATION

Each student has to submit two copies of the dissertation work duly forwarded by the HOD of Department concerned. The forwarded copies will be submitted in the Department of Botany, RKDF University, for evaluation (Seven days before the seminar).

The paper will consist of

- (a) Field work/Lab work related to the project.
 - (b) Preparation of dissertation based on the work undertaken.
 - (c) Presentation of project work in the seminar on the assigned topic in the P.G. Department of Botany, RKDF University, Ranchi & open viva thereon.
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- ☐ Student alone or in a group of not more than five, shall undertake one Project approved by the Subject Teacher/H.O.D. of the Department/College concerned. The progress of the Project shall be monitored by the faculty members at regular intervals.
- OR
- ☐ Paper presentation on 'Topic Provided' and group discussion